Practical Training Report: WPA Network Penetration Testing Using a Dictionary Attack

Student Information:

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1. Objective

The objective of this exercise is to demonstrate the ability to analyze a packet capture file containing a WPA handshake and execute a dictionary attack to extract the network's encryption key (password).

2. Tools and Environment

The following tools and files were utilized during the execution of this task:

- **Operating System:** Linux environment (specifically Kali Linux, indicated by the kali user prompt).
- **Primary Tool:** aircrack-ng (Version 1.7).
- **Capture File:** wpa.full.cap (contains the captured handshake data).
- **Wordlist:** The standard rockyou.txt dictionary file, located at /usr/share/wordlists/rockyou.txt.

3. Methodology and Execution

The attack was executed via the command-line interface with root privileges (sudo) to ensure the tool had the necessary read access to the files. The command executed was:

Bash

```
sudo aircrack-ng wpa.full.cap -w /usr/share/wordlists/rockyou.txt
```

Command Breakdown:

- aircrack-ng: Invokes the cracking utility.
- wpa.full.cap: Specifies the target capture file to be analyzed.
- -w /usr/share/wordlists/rockyou.txt: Defines the path to the wordlist file used for testing potential passwords.

4. Analysis of Results

The terminal output indicates the tool progressed through the following steps and extracted the corresponding data:

- **Data Reading:** The tool read 15 packets from the wpa.full.cap file.
- **Target Identification:** The tool detected one valid WPA handshake and identified the target network with the following parameters:
    - **BSSID (MAC Address):** 00:14:6C:7E:40:80
    - **ESSID (Network Name):** teddy
    - **Encryption Type:** WPA
- **Cracking Process:** The network was automatically selected as the target ("Choosing first network as target"). The tool began testing keys at a rate of approximately **4058.01 keys/second** (k/s).
- **Final Result:** A match was successfully found.
    - **The extracted encryption key is:** 44445555

5. Conclusion and Security Recommendations

This practical exercise proves that WPA encryption protocols are vulnerable to dictionary attacks when weak, predictable, or commonly used passwords are in place.

Mitigation Recommendations:

1. Avoid using short passwords or those relying on repetitive numeric patterns (e.g., 44445555).
2. Implement complex passwords exceeding 12 characters, utilizing a random mix of uppercase letters, lowercase letters, numbers, and symbols.
3. Upgrade the network encryption protocol to WPA3 where supported, as it provides robust protection against offline dictionary attacks regardless of password complexity.

```
└─$ sudo aircrack-ng wpa.full.cap -w /usr/share/wordlists/rockyou.txt
[sudo] password for kali:
Reading packets, please wait...
Opening wpa.full.cap
Read 15 packets.

   #  BSSID              ESSID                     Encryption

   1  00:14:6C:7E:40:80  teddy                     WPA (1 handshake)

Choosing first network as target.

Reading packets, please wait...
Opening wpa.full.cap
Read 15 packets.

1 potential targets


                               Aircrack-ng 1.7

      [00:00:20] 79910/14344392 keys tested (4058.01 k/s)

      Time left: 58 minutes, 35 seconds                          0.56%

                              KEY FOUND! [ 44445555 ]
```